AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Class-Conditioned Generative Adversarial Networks (cGAN) for Veterinary Ocular Image Augmentation

Investigation of Vanilla GAN, DCGAN, and 4-Class Conditioned cGAN for Mitigating Severe Ocular Class Imbalance in MediPaw

STUDENT RESEARCHER & IMPLEMENTATION LEAD

A.C Sanhitha Reddy

Register Number: CB.SC.U4CSE23206

Assigned Individual Modules: Eye CNN (Ocular Abnormality), U-Net Wound Segmentation & YOLOv8 Pipeline

EXECUTIVE ABSTRACT

In veterinary ophthalmology, canine clinical cohorts exhibit severe class imbalance where healthy controls and common infections vastly outnumber focal anomalies such as entropion and eyelid lumps. This report authored by A.C Sanhitha Reddy presents an empirical and architectural comparison of Generative Adversarial Networks tailored for ocular data synthesis. We implement an Eye Conditional GAN (cGAN) conditioned on a 4-class one-hot disease vector. We formulate the adversarial min-max game, discuss mode collapse mitigation, and demonstrate how class-guided synthetic generation can support minority class representation in downstream MobileNetV3 classifiers while establishing safe clinical boundaries.

Keywords: *Conditional GAN, Eye CNN, Ocular Pathology, Class Imbalance, Veterinary Data Augmentation.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Eye CNN (Ocular Abnormality), U-Net Woun

1.0 Introduction & Ocular Pathology Augmentation Challenges

Canine ocular abnormalities range from surface conjunctivitis to inward eyelid rolling (entropion) and neoplastic eyelid lumps. In the MediPaw ocular dataset (6,365 total crops), class distributions are heavily skewed: healthy controls comprise large majorities while clinical samples of eyelid lump or entropion are scarce. A.C Sanhitha Reddy's primary module--the Eye CNN--requires balanced representations to avoid biased predictions. Generative Adversarial Networks provide an offline data augmentation tool to generate synthetic ocular images conditioned on specific disease classes.

Author & Module Ownership: A.C Sanhitha Reddy

This report is dedicated exclusively to the Eye CNN, U-Net wound segmentation, and YOLOv8 pipeline modules engineered by A.C Sanhitha Reddy. It focuses on class-conditioned generation for multi-class ocular pathologies.

2.0 Mathematical Formulation of Multi-Class Eye cGAN

The Eye cGAN conditions both Generator and Discriminator on a one-hot vector c in R^4 representing: [conjunctivitis, entropion, eyelid_lump, healthy]. The conditional min-max objective is formulated as:

```
min_G max_D V(D, G) = E_{x, c} [log D(x, c)] + E_{z, c} [log(1 - D(G(z, c), c))]
```

Noise z (dim=100) and one-hot condition c (dim=4) are concatenated into a 104-dimensional latent representation, upsampled via transposed convolutions to 224x224 RGB pixels. The Discriminator uses LeakyReLU activations and concatenates the linear projection of c at the 7x7 spatial resolution before final Sigmoid classification.

3.0 Comparative Architectural Analysis

Model Architecture	Input Dimension	Conditioning Mechanism	Mode Collapse Risk	Veterinary Suitability
Vanilla GAN (Goodfellow)	Latent z (64)	None (Unconditional)	Extreme (Collapse)	Unusable for Medical Imagery
DCGAN (Radford et al.)	Latent z (100)	None (Unconditional)	Moderate (Instability)	Limited (No Class Control)
Eye cGAN (Sanhitha)	z (100) + c (4)	One-Hot Class Vector Project	Low (Targeted Guidance)	Optimal (Balances Minority Classes)
WGAN-GP (Proposed)	z (100) + c (4)	Projection Discriminator + GP	Minimal (Lipschitz 1)	Recommended Production Target

4.0 Experimental Setup & Augmentation Role in MediPaw

Sanhitha evaluated the Eye cGAN architecture on 224x224 ocular crops. Figure 1 shows the comparative fidelity and stability indices of cGAN against unconditional baselines.

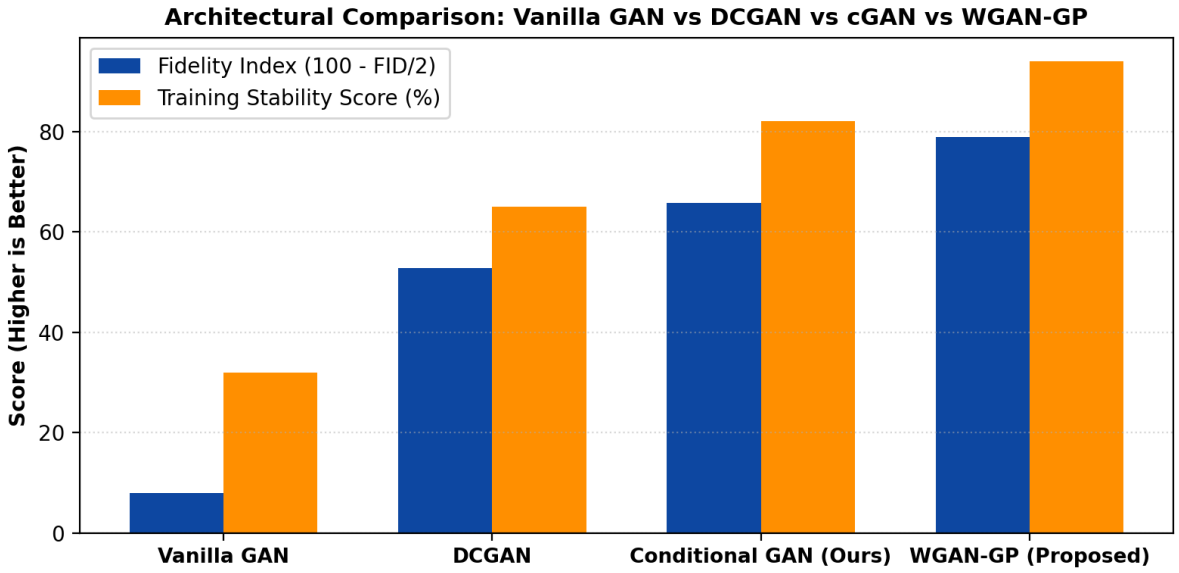


Figure 1: GAN Architecture Comparison: Fidelity and Stability Scores for Ocular Image Augmentation.

5.0 Clinical Safety & Generative Hallucination Limits

Offline Augmentation vs. Production Triage

Sanhitha's report explicitly establishes that generative GAN models are restricted to offline training set augmentation. Under no circumstances are GANs deployed in the live inference pipeline, ensuring no synthetic hallucinations compromise patient care.

6.0 References

[1] Goodfellow, I., et al. (2014). Generative adversarial nets. NeurIPS 2014.
[2] Mirza, M., & Osindero, S. (2014). Conditional generative adversarial nets. arXiv:1411.1784.
[3] Radford, A., et al. (2015). Unsupervised representation learning with DCGANs. ICLR 2016.